GAME & SIMULATION DESIGN FOR SPACE APPS

LEAVE WITH A PLAYABLE SPEC.

A 2-hour workshop · NASA Space Apps Cairo 2026

MARWAN AYYAD

Your host for the next 120 minutes

HELLO. I'M MARWAN, AND HERE'S OUR NEXT 2 HOURS.

By the end, your team has a one-minute game spec ready to build.

10	20	15	15	15	35	10
Why games, not dashboards	The core loop	How winners did it	The 48-hour playbook	Live demo	You build a spec	Rules & questions

IF THE PLAYER HAS NO MEANINGFUL CHOICE, IT'S NOT A GAME.

IT'S JUST A CHORE.

SAME MARS DATA. TWO EXPERIENCES.

A DASHBOARD

−63°C Outside temperature	**2.1** Dust opacity (τ)
38% Solar output	**6 sols** Oxygen reserve

You read it. **Nothing you do changes anything.**

Teaching numbers, not live data.

A SIMULATION: 3 SOLS OF STORM

Power		60
Warmth		60
Oxygen		60

Each sol, the storm drains warmth and oxygen. Pick one move:

Heat the habitat | **Run oxygen plant**

Power down | **Reset**

WHY A CHOICE BEATS A CHART

FAST FEEDBACK

Every move answers back within a second.

FAILURE IS SAFE

Crash the habitat, press reset, try a smarter plan.

DECISIONS STICK

People remember what they chose, not what they scrolled past.

EVERY GAME RUNS ONE LOOP

ACTION

The player makes a choice.

→

RESPONSE

The rules change the world's state.

→

FEEDBACK

The player sees what happened, at once.

↵ …and the feedback sets up the next choice.

STEP 1 • ACTION: GIVE THE PLAYER A DILEMMA

NO DILEMMA

"Press **Launch** to launch the rocket."

There's only one sensible move, so there's nothing to decide.

A REAL TRADE-OFF

"Launch now in bad weather, or wait a day and lose the orbital window?"

Both options cost something. **That's a choice.**

STEP 2 • RESPONSE: RULES ANYONE CAN READ

A RULE IS ONE LINE OF CAUSE AND EFFECT

Same choice, same result, every time. No dice, no black-box AI.

If the player can't predict the rule, they can't plan, so they can't really choose.

RULE CARD

- Heaters on → power −25, warmth +25
- Every sol in the storm → warmth −20, oxygen −20
- Any meter hits 0 → mission over

STEP 3 • FEEDBACK: SHOW IT WITHIN A SECOND

COLOUR

Green is safe. Yellow is a warning. Red is trouble. No reading needed.

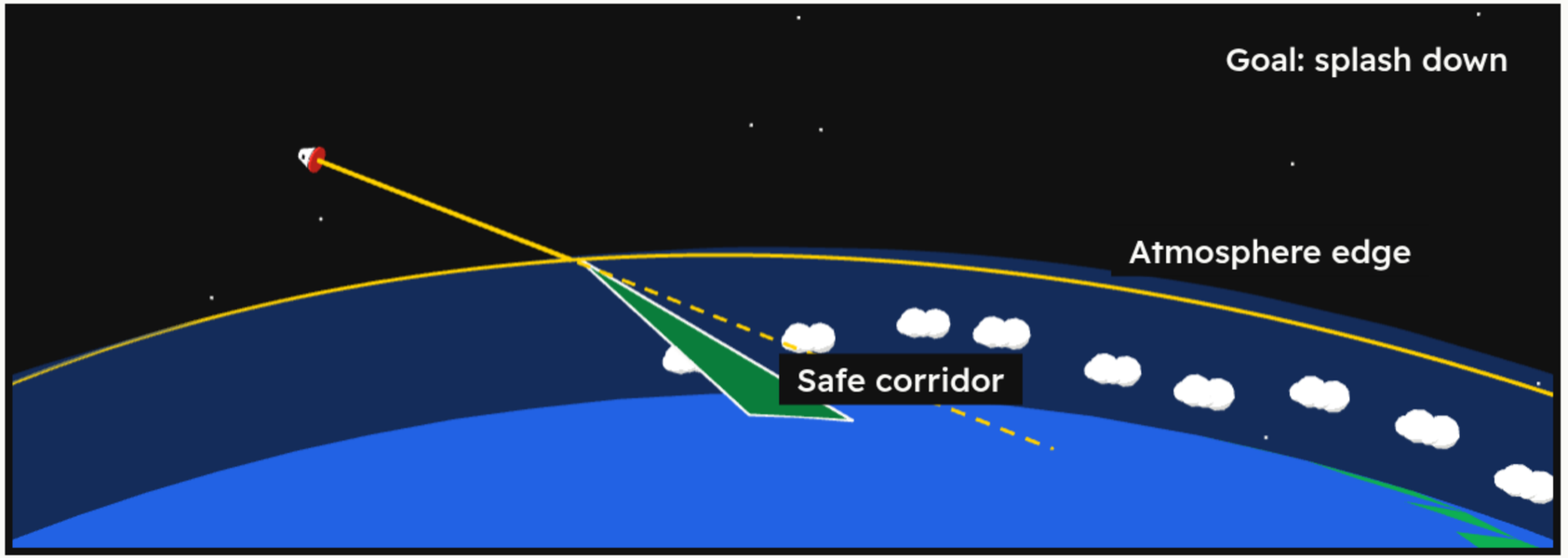
SOUND

A beep on every change, an alarm before failure.

NUMBERS

Show the exact change: "oxygen −20". Players learn the rules from it.

TRY IT: ONE DIAL, ONE LOOP



Capsule entry angle

−4.0°

Commit

Drawn 6× steeper. Drag to turn.

ACTION

Pick an angle and commit.

RESPONSE

The rules will run here.

FEEDBACK

You'll see the outcome here.

Simplified corridor: −5.5° to −7.5°.

TURN A CHALLENGE INTO VARIABLES

THE CHALLENGE TEXT

"Design a base that keeps a crew alive at the lunar south pole: power is limited during the long night, temperatures drop sharply, and oxygen must be produced on site."

Example written for this workshop.

WHAT TO PULL OUT

- **Resources** that run out: power, oxygen
- **Threats** that push back: cold, the long night
- **The goal**: survive until sunrise

KEEP IT TO 2–4 VARIABLES

1 VARIABLE

Too shallow. There's nothing to trade off, so it's a quiz.

3 VARIABLES

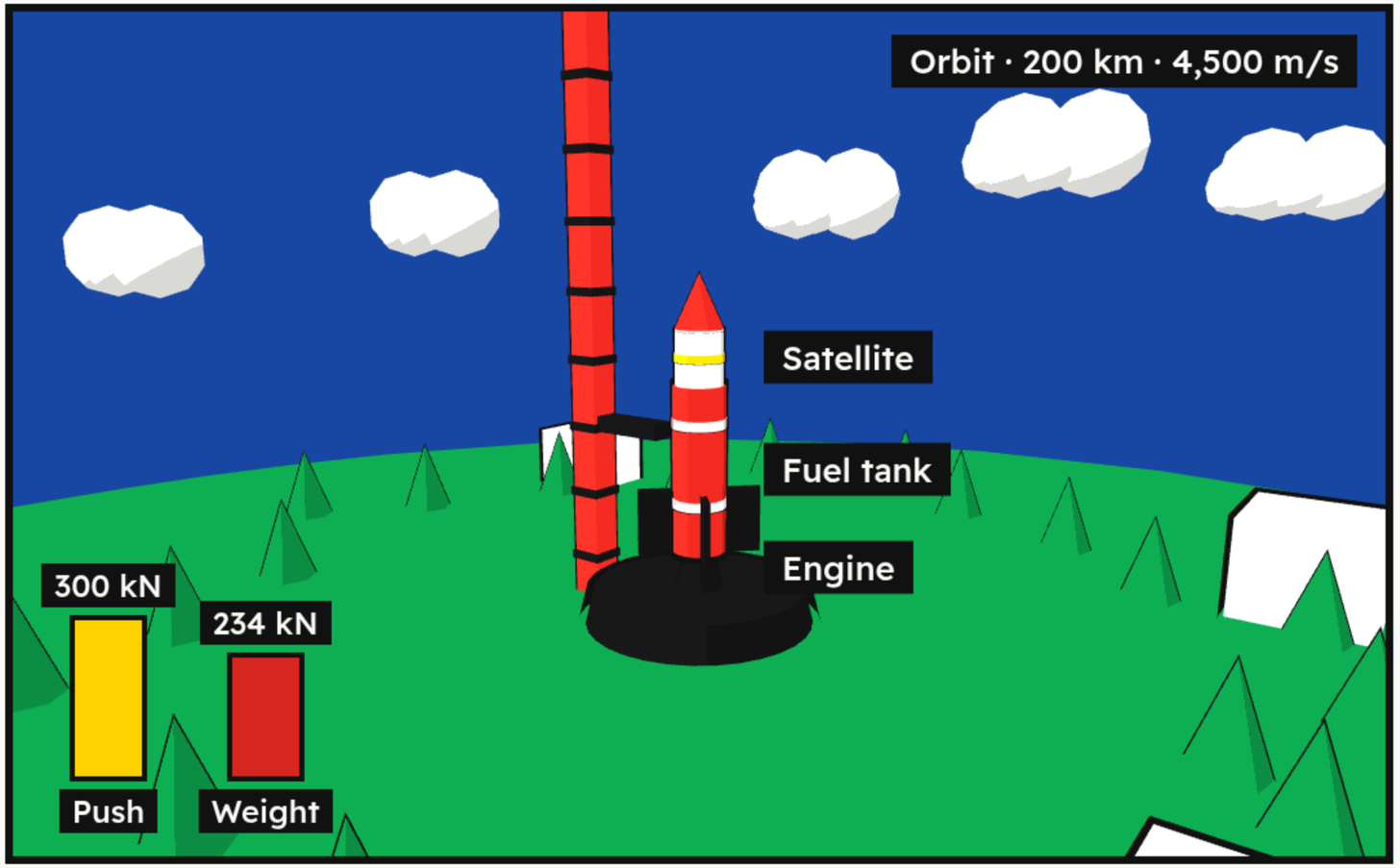
The sweet spot. Enough tension for real choices, small enough to finish in 48 hours.

5+ VARIABLES

Scope death. You'll spend Saturday balancing and ship nothing Sunday.

TOO HEAVY TO FLY?

Get the heaviest satellite to orbit. Too little fuel: you fall back. Too much: stuck on the pad.



Satellite (heavier = more stars) 2,000 kg

Fuel (+8% tank weight) 20,000 kg

Engine push (bigger = heavier) 300 kN

Launch

READY?

Set the sliders, press Launch.

Real rules, teaching numbers.

CLIMB THE FIDELITY LADDER

RAW SCIENCE

Datasets and papers. True, but no player can act on them.

FULL PHYSICS

Exact equations. The 48-hour trap: you'll debug maths all weekend.

GAME RULES

The science simplified into clear states: safe, warning, critical. **Build here.**

ZUMORRODA-X

Scan to play their game, Henry's Farm Adventures

Egypt · 2025
Global winner, Art & Technology Award
Challenge: NASA Farm Navigators

WHAT THEY BUILT

Mini games where you play a farmer on a journey around the world, adapting to heat waves, flooding and other environmental challenges.

STEAL THIS

Put the player in someone's shoes. The data arrives as events they must react to.

Sources: NASA, 2025 Space Apps global winners announcement; the team's Space Apps project page.

TERRATALES

Scan for their project, demo video and site

Cairo, Egypt · 2024
Global winner, Best Storytelling Award

WHAT THEY BUILT

An interactive game that encourages players to make eco-friendly choices, showing the climate outcome of acting or not acting.

STEAL THIS

Let each choice show its consequence. The story is what the player caused.

Sources: NASA, 2024 global winners announcement; Universe Magazine; the team's Space Apps project page.

ECO-METROPOLIS

Scan for their project, demo video and code

Team Innovisionaries
Sharjah, UAE · 2024
Global winner, Most Inspirational Award

WHAT THEY BUILT

A city-building game where players make urban-planning and resource-management decisions.

STEAL THIS

Scarce resources force real trade-offs. Give the player less than they want.

Sources: NASA, 2024 Space Apps global winners announcement; the team's Space Apps project page.

PUREFLOW

Scan to try PureFlow

Brazil · 2025
Global winner, Best Mission Concept Award

WHAT THEY BUILT

A platform to design and model a space habitat in 3D, then test it against space-weather threats like solar storms.

STEAL THIS

A simulation can be a test bench: build it, stress it, see what breaks.

Sources: NASA, 2025 Space Apps global winners announcement; the team's Space Apps project page.

THREE MUST-HAVES. THEN ONE EXTRA.

EVERY GAME NEEDS

- **Choices:** a real dilemma
- **Challenge:** never enough to go around, like Eco-Metropolis
- **Feedback:** see what broke, like PureFlow

PICK ONE EXTRA

- **Story**, like TerraTales
- **Characters**, like Henry in Zumorroda-X
- **Mystery:** data reveals a clue
- **Puzzle:** a combination to crack

FUN IS THE RESULT, NOT AN INGREDIENT.

EGYPT HAS DONE THIS TWICE. WITH GAMES.

2 OF 20

Global winners in 2024–2025 that were Egyptian game teams: TerraTales and Zumorroda-X.

2025's ten winners were picked from 18,860 teams. Source: NASA.

WHAT THE WINNERS SHARE

- The player makes a **decision**
- NASA data lives **inside the rules**
- One complete loop, **finished**

NEXT COULD BE YOUR TEAM.

FOUR IDEAS THAT KILL A 48-HOUR PROJECT



INSTEAD: ONE SCREEN, ONE LOOP, ONE FINISHED LEVEL.

USE THE TOOL YOU ALREADY KNOW

GOOD PICKS FOR 48 HOURS

- **Browser** (HTML + canvas): judges click a link and play
- **Godot**: light, free, fast 2D
- **Three.js**: 3D in the browser, like the capsule and rocket scenes in this deck
- **Unity**: only if your team already knows it

ONE RULE: TRACK YOUR VARIABLES IN ONE SHARED DOC.

Every number and every rule, in one place the whole team can edit.

AI: A FAST COPILOT, A BAD SCIENTIST

USE AI FOR

- Writing state and rule code
- Balancing numbers so it's not too easy
- Placeholder art and sounds

NEVER FOR

- Inventing the science
- Numbers you can't trace to a NASA source

Check every number it gives you.

GOT AN AI AGENT?

Claude Code, Cursor, Antigravity, Codex: it opens your project, runs the game, reads the error and fixes it. You still review every change it makes.

SPEC IN, PLAYABLE DRAFT OUT

THE AI PROTOTYPE LOOP

- **Paste** your spec after the prompt
- **Save** the reply as game.html, double-click it
- **Play** one minute: does the choice matter?
- **Ask** for one change at a time; paste errors exactly
- **Keep** every version that works

PROMPT STARTER

> "Build one HTML file that runs offline. Use only the rules in my spec; ask before inventing numbers. One button per decision, green-yellow-red meters, show every change in numbers."

Scan for the full prompt

Copy full prompt

Never paste passwords, keys or personal details.
Follow your event's AI rules and say what AI made.

THE RECIPE, ON A MARS ROVER

1 • GOAL

Send 3 rock samples home before the dust storm hits.

2 • VARIABLES

Battery, dust on the solar panels, samples.

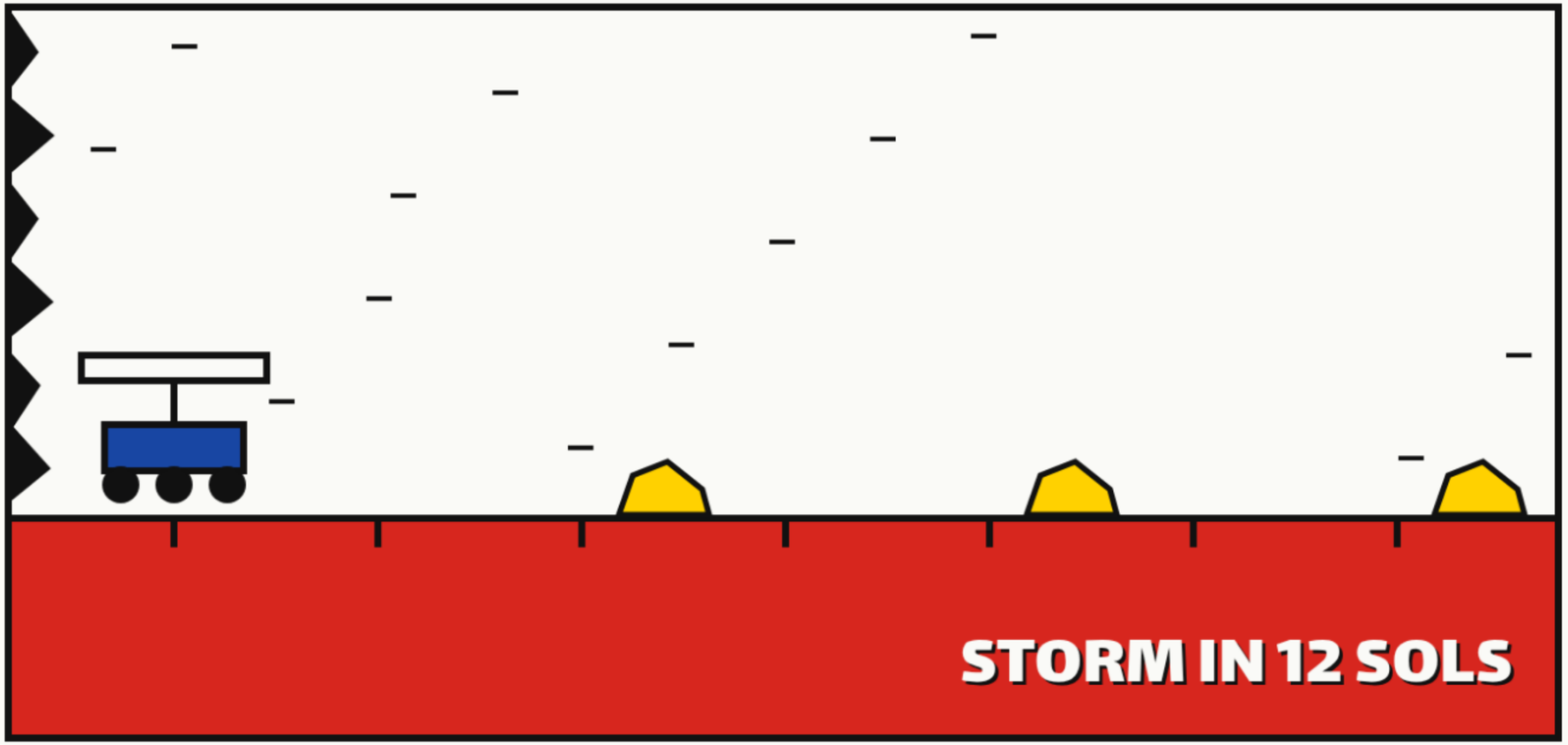
3 • DECISION

Drive, drill a rock, or spend a day cleaning the panels?

4 • FEEDBACK

Meters repaint, dust piles up, the storm creeps closer.

THREE ROCKS HOME BEFORE THE STORM



1 / 12
Sol · storm in 12

100%
Battery

0%
Dust on panels

0 / 3
Rock samples

Teaching numbers, not mission data.

1 · Drive | **2 · Sample** | **3 · Clean panels**

Sol 1. Vote each sol: 1 drive, 2 sample a rock, 3 clean the panels.

Reset

YOUR TURN: DESIGN A ONE-MINUTE GAME

Pick any NASA challenge. Your team writes one spec. Tonight, AI turns it into a playable draft.

20 MIN

DESIGN

Fill in the template on the next slide.

10 MIN

SWAP

Trade specs with the next team. They play your rules on paper.

5 MIN

REVISE

Fix the one rule they found confusing.

THE ONE-MINUTE SPEC

1 · GOAL (ONE SENTENCE)

Keep the lunar crew alive through a 14-day night. One action per day.

2 · THREE VARIABLES

- Battery: starts 100 kWh, −2 per day
- Heat: starts +10 °C, −6 per day, safe above −10
- Oxygen: starts 80%, −10% per day

3 · TWO DECISIONS

- Run heaters: heat +12, battery −8
- Run oxygen recycler: oxygen +15, battery −8

4 · RULE TABLE

- Battery 0 → machines off, heat −10 per day
- Heat below −20 for 3 days → mission lost
- Oxygen 0 → mission lost

5 · END STATE

Survive day 14 → crew survives.

Copy blank template

Copy as AI prompt

Scan for the full prompt

HANDS ON LAPTOPS

20:00

DESIGN

Start | **Design 20** | **Swap 10** | **Revise 5**

DONE = ONE LEVEL THAT PLAYS. GRAPHICS LAST.

THE 5 GOLDEN RULES

- Design the decisions first
- Put the science inside the rules
- Give feedback within a second
- Simplify the science, openly
- Ship one complete loop

ASK ME ANYTHING

My links

Join the Discord

discord.gg/fycj8nKgKE
marwan0.github.io
marwan_ayad@live.com
LinkedIn: in/marwan-ayyad
YouTube: @Marwan-Tech
Facebook: @marwan.ayyad.7

BLOOD TAX

Scan to watch how to play

Mera Mera Games
Built solo in Unity
Aiming for a Steam Next Fest demo

ONE MORE THING: MY GAME

One mosquito against the humans.

- **Mosquito:** follow heat through walls, hide in plain sight, bite, fly the blood to a jar on the balcony.
- **Humans:** swat, spray, clap, listen for the buzz.

YOUR TURN

Watch it, then tell me one thing you'd change.